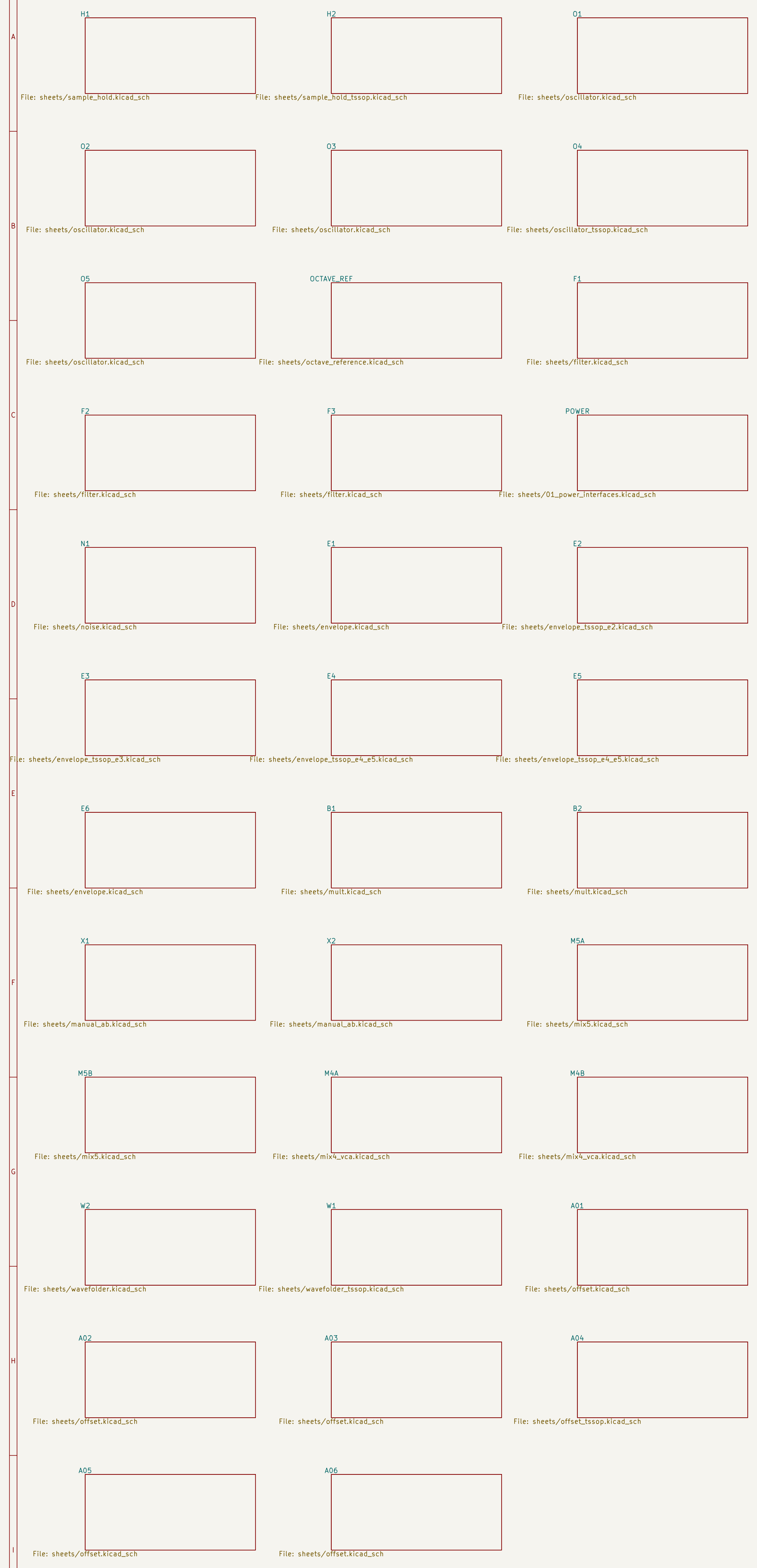
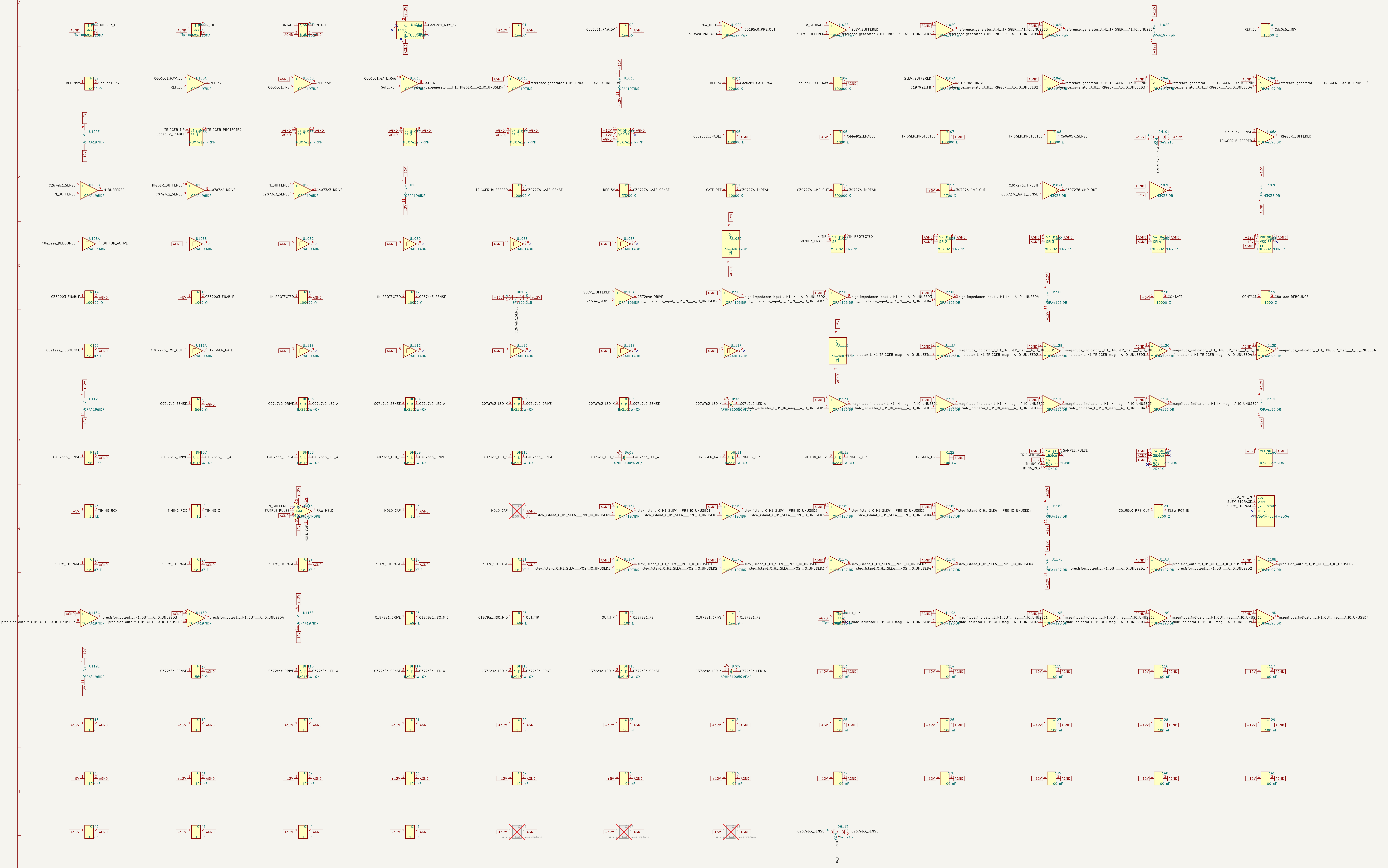

H1
File: sheets/sample_hold.kicad_sch
H2
File: sheets/sample_hold_tssop.kicad_sch
O1
File: sheets/oscillator.kicad_sch
O2
File: sheets/oscillator.kicad_sch
O3
File: sheets/oscillator.kicad_sch
O4
File: sheets/oscillator_tssop.kicad_sch
O5
File: sheets/oscillator.kicad_sch
OCTAVE_REF
File: sheets/octave_reference.kicad_sch
F1
File: sheets/filter.kicad_sch
F2
File: sheets/filter.kicad_sch
F3
File: sheets/filter.kicad_sch
POWER
File: sheets/01_power_interfaces.kicad_sch
N1
File: sheets/noise.kicad_sch
E1
File: sheets/envelope.kicad_sch
E2
File: sheets/envelope_tssop_e2.kicad_sch
E3
File: sheets/envelope_tssop_e3.kicad_sch
E4
File: sheets/envelope_tssop_e4_e5.kicad_sch
E5
File: sheets/envelope_tssop_e4_e5.kicad_sch
E6
File: sheets/envelope.kicad_sch
B1
File: sheets/mult.kicad_sch
B2
File: sheets/mult.kicad_sch
X1
File: sheets/manual_ab.kicad_sch
X2
File: sheets/manual_ab.kicad_sch
M5A
File: sheets/mix5.kicad_sch
M5B
File: sheets/mix5.kicad_sch
M4A
File: sheets/mix4_vca.kicad_sch
M4B
File: sheets/mix4_vca.kicad_sch
W2
File: sheets/wavefolder.kicad_sch
W1
File: sheets/wavefolder_tssop.kicad_sch
A01
File: sheets/offset.kicad_sch
A02
File: sheets/offset.kicad_sch
A03
File: sheets/offset.kicad_sch
A04
File: sheets/offset_tssop.kicad_sch
A05
File: sheets/offset.kicad_sch
A06
File: sheets/offset.kicad_sch

Sheet: /
File: zudo–osc–hole–field.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 1/36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RAW_TIP
CONTACT
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REF_N5V
Cdc0c61_INV
REF_5V
Cdc0c61_GATE_RAW
GATE_REF
RAW_HELD
C5195c0_PRE_OUT
SLEW_STORAGE
SLEW_BUFFERED
SLEW_POT_IN
TRIGGER_PROTECTED
Cdded02_ENABLE
Ce0e057_SENSE
TRIGGER_BUFFERED
C267eb3_SENSE
IN_BUFFERED
C07a7c2_DRIVE
C07a7c2_SENSE
C07a7c2_LED_A
Ca073c3_DRIVE
Ca073c3_SENSE
Ca073c3_LED_A
C307276_GATE_SENSE
C307276_THRESH
C307276_CMP_OUT
C8a1aae_DEBOUNCE
BUTTON_ACTIVE
C382003_ENABLE
IN_PROTECTED
IN_TIP
TRIGGER_GATE
TRIGGER_OR
SAMPLE_PULSE
TIMING_RCX
TIMING_C
HOLD_CAP
C1979a1_DRIVE
C1979a1_FB
C1979a1_ISO_MID
OUT_TIP
C372c4e_DRIVE
C372c4e_SENSE
C372c4e_LED_A
OPA4197IDR
OPA4196IDR
SN74HC14DR
TMUX7412FRRPR
CD74HC221M96
BAS16GW-QX
APHHS1005QWF/D
BAT54S,215
+12V
-12V
+5V
AGND

Sheet: /H1/
File: sample_hold.kicad_sch

Title:

Size: A1 Date:
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Id: 2/36

TRIGGER_TIP
IN_TIP
CONTACT
Cdc0c61_RAW_5V
Cdc0c61_INV
REF_5V
REF_N5V
RAW_HELD
C5195c0_PRE_OUT
SLEW_STORAGE
SLEW_BUFFERED
Cdc0c61_GATE_RAW
GATE_REF
C1979a1_DRIVE
C1979a1_FB
TRIGGER_PROTECTED
Cdded02_ENABLE
Ce0e057_SENSE
TRIGGER_BUFFERED
C267eb3_SENSE
IN_BUFFERED
C07a7c2_DRIVE
C07a7c2_SENSE
Ca073c3_DRIVE
Ca073c3_SENSE
C307276_GATE_SENSE
C307276_THRESH
C307276_CMP_OUT
C8a1aae_DEBOUNCE
BUTTON_ACTIVE
C382003_ENABLE
IN_PROTECTED
C372c4e_DRIVE
C372c4e_SENSE
TRIGGER_GATE
C07a7c2_LED_A
C07a7c2_LED_K
Ca073c3_LED_A
Ca073c3_LED_K
TRIGGER_OR
SAMPLE_PULSE
TIMING_RCX
TIMING_C
HOLD_CAP
SLEW_POT_IN
C1979a1_ISO_MID
OUT_TIP
C372c4e_LED_A
C372c4e_LED_K
reference_generator_J_H1_TRIGGER___A1_IO_UNUSED3
high_impedance_input_J_H1_IN___A_IO_UNUSED1
magnitude_indicator_J_H1_TRIGGER_mag___A_IO_UNUSED1
magnitude_indicator_J_H1_IN_mag___A_IO_UNUSED1
slew_island_C_H1_SLEW___PRE_IO_UNUSED1
slew_island_C_H1_SLEW___POST_IO_UNUSED1
precision_output_J_H1_OUT___A_IO_UNUSED1
magnitude_indicator_J_H1_OUT_mag___A_IO_UNUSED1
OPA4197IPWR
OPA4197IDR
OPA4196IDR
SN74HC14DR
TMUX7412FRRPR
LM393BIDR
BAS16GW-QX
APHHS1005QWF/D
BAT54S,215
BAV199,215
CD74HC221M96
AGND
+12V
-12V
+5V
DNP
Reservation
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
A
B
C
D
E
F
G
H
I
J
K

Sheet: /H2/
File: sample_hold_tssop.kicad_sch

Title:

Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 3/36

Sheet: /01/
File: oscillator.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/36

Sheet: /02/
File: oscillator.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 5/36

Sheet: /03/
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Title:
Size: A0 Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 6/36

Sheet: /04/
File: oscillator_tssop.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/36

Sheet: /05/
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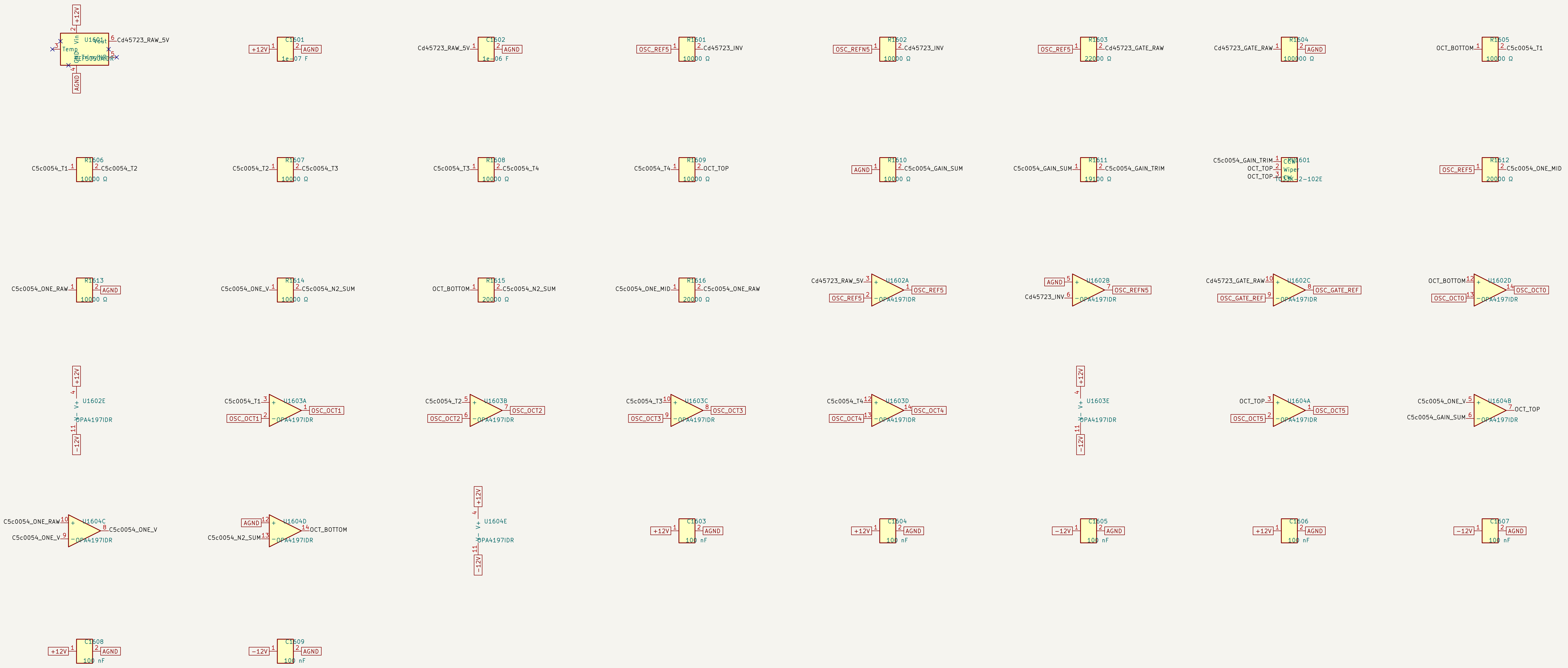
Title:
Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 8/36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REF5050AIDR
+12V
2 Vin
3 Temp
4 GND
AGND
6 Vout
Cd45723_RAW_5V
5 Trim/NR
C1601
+12V
AGND
1e-07 F
C1602
Cd45723_RAW_5V
AGND
1e-06 F
R1601
OSC_REF5
Cd45723_INV
10000 Ω
R1602
OSC_REFN5
Cd45723_INV
10000 Ω
R1603
OSC_REF5
Cd45723_GATE_RAW
22000 Ω
R1604
Cd45723_GATE_RAW
AGND
100000 Ω
R1605
OCT_BOTTOM
C5c0054_T1
10000 Ω
R1606
C5c0054_T1
C5c0054_T2
10000 Ω
R1607
C5c0054_T2
C5c0054_T3
10000 Ω
R1608
C5c0054_T3
C5c0054_T4
10000 Ω
R1609
C5c0054_T4
OCT_TOP
10000 Ω
R1610
AGND
C5c0054_GAIN_SUM
10000 Ω
R1611
C5c0054_GAIN_SUM
C5c0054_GAIN_TRIM
19100 Ω
RV1601
TC33X-2-102E
C5c0054_GAIN_TRIM
OCT_TOP
OCT_TOP
CCW
Wiper
CW
R1612
OSC_REF5
C5c0054_ONE_MID
20000 Ω
R1613
C5c0054_ONE_RAW
AGND
10000 Ω
R1614
C5c0054_ONE_V
C5c0054_N2_SUM
10000 Ω
R1615
OCT_BOTTOM
C5c0054_N2_SUM
20000 Ω
R1616
C5c0054_ONE_MID
C5c0054_ONE_RAW
20000 Ω
U1602A
OPA4197IDR
Cd45723_RAW_5V
OSC_REF5
OSC_REF5
U1602B
OPA4197IDR
AGND
Cd45723_INV
OSC_REFN5
U1602C
OPA4197IDR
Cd45723_GATE_RAW
OSC_GATE_REF
OSC_GATE_REF
U1602D
OPA4197IDR
OCT_BOTTOM
OSC_OCT0
OSC_OCT0
U1602E
OPA4197IDR
+12V
−12V
U1603A
OPA4197IDR
C5c0054_T1
OSC_OCT1
OSC_OCT1
U1603B
OPA4197IDR
C5c0054_T2
OSC_OCT2
OSC_OCT2
U1603C
OPA4197IDR
C5c0054_T3
OSC_OCT3
OSC_OCT3
U1603D
OPA4197IDR
C5c0054_T4
OSC_OCT4
OSC_OCT4
U1603E
OPA4197IDR
+12V
−12V
U1604A
OPA4197IDR
OCT_TOP
OSC_OCT5
OSC_OCT5
U1604B
OPA4197IDR
C5c0054_ONE_V
C5c0054_GAIN_SUM
OCT_TOP
U1604C
OPA4197IDR
C5c0054_ONE_RAW
C5c0054_ONE_V
C5c0054_ONE_V
U1604D
OPA4197IDR
AGND
C5c0054_N2_SUM
OCT_BOTTOM
U1604E
OPA4197IDR
+12V
−12V
C1603
+12V
AGND
100 nF
C1604
+12V
AGND
100 nF
C1605
−12V
AGND
100 nF
C1606
+12V
AGND
100 nF
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AGND
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C1608
+12V
AGND
100 nF
C1609
−12V
AGND
100 nF

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Title:

Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 9/36

Sheet: /F1/
File: filter.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 10/36

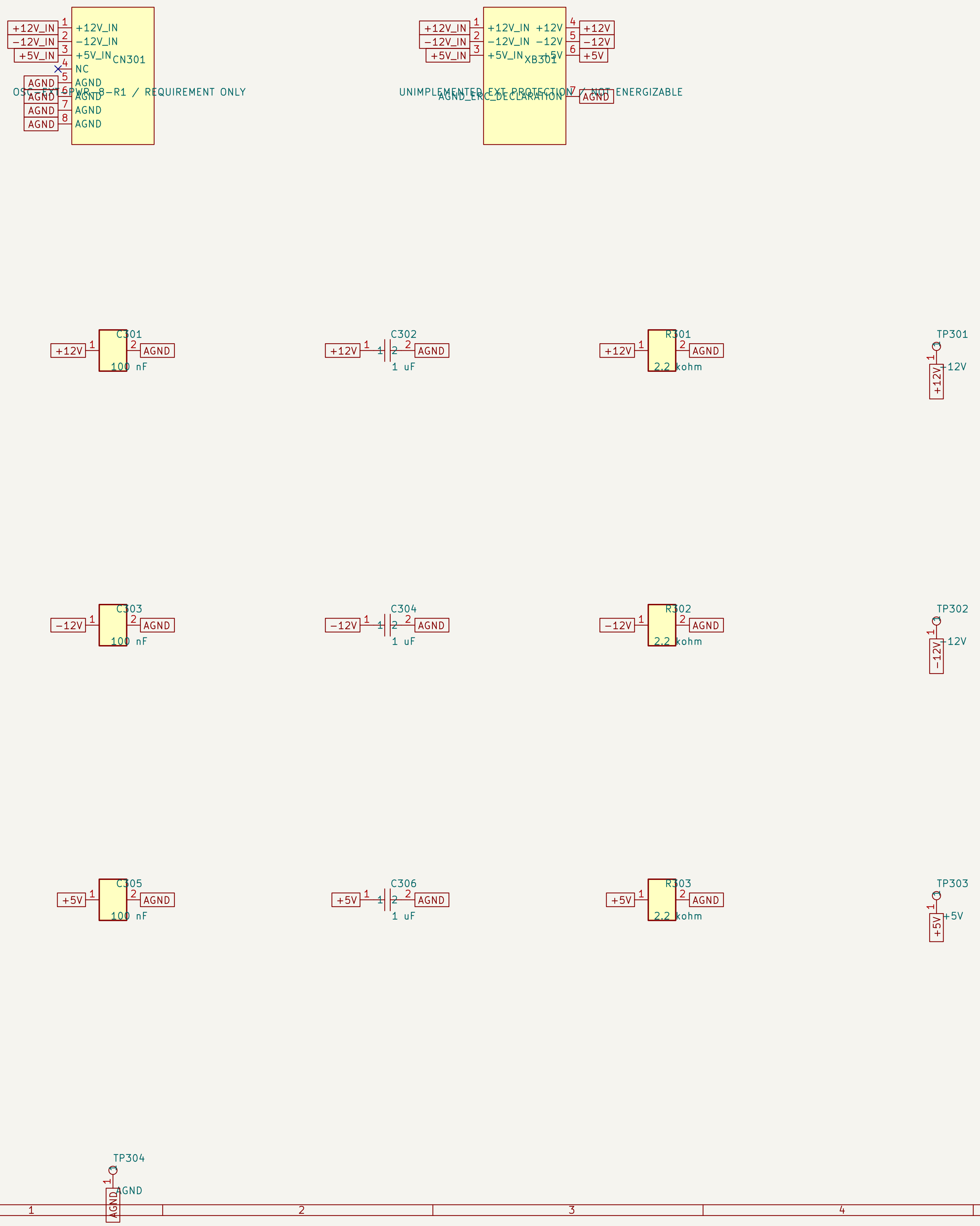
Sheet: /F2/
File: filter.kicad_sch

Title:

Size: A0 Date:
KiCad E.D.A. 10.0.6

Rev:
Id: 11/36

Sheet: /F3/
File: filter.kicad_sch
Title:
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 12/36

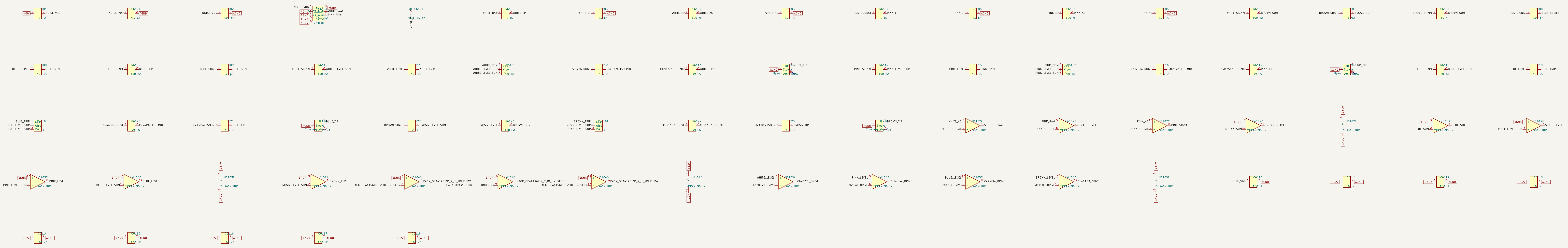

CN301
+12V_IN
−12V_IN
+5V_IN
NC
AGND
1
2
3
4
5
6
7
8
OSC-EXT-PWR-8-R1 / REQUIREMENT ONLY
XB301
+12V_IN +12V
−12V_IN −12V
+5V_IN +5V
AGND_ERC_DECLARATION
+12V
−12V
+5V
UNIMPLEMENTED EXT PROTECTION / NOT ENERGIZABLE
C301
100 nF
C302
1 uF
R301
2.2 kohm
TP301
C303
100 nF
C304
1 uF
R302
2.2 kohm
TP302
C305
100 nF
C306
1 uF
R303
2.2 kohm
TP303
TP304

Sheet: /POWER/
File: 01_power_interfaces.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 13/36


R6101 +5V NOISE_VDD 22 Ω
C6101 NOISE_VDD AGND 4.7 µF
C6102 NOISE_VDD AGND 100 nF
U6101 NOISE_VDD +5V AGND WHITE_RAW PINK_RAW NOISE2 Unused
#FLG6101 NOISE_VDD FILTERED_5V
R6102 WHITE_RAW WHITE_LP 1 kΩ
C6103 WHITE_LP AGND 10 nF
C6104 WHITE_LP WHITE_AC 100 nF
R6103 WHITE_AC AGND 100 kΩ
R6104 PINK_SOURCE PINK_LP 1 kΩ
C6105 PINK_LP AGND 10 nF
C6106 PINK_LP PINK_AC 100 nF
R6105 PINK_AC AGND 100 kΩ
R6106 WHITE_SIGNAL BROWN_SUM 100 kΩ
R6107 BROWN_SHAPE BROWN_SUM 1 MΩ
C6107 BROWN_SHAPE BROWN_SUM 10 nF
C6108 PINK_SIGNAL BLUE_SERIES 100 pF
R6108 BLUE_SERIES BLUE_SUM 100 kΩ
R6109 BLUE_SHAPE BLUE_SUM 100 kΩ
C6109 BLUE_SHAPE BLUE_SUM 33 pF
R6110 WHITE_SIGNAL WHITE_LEVEL_SUM 100 kΩ
R6111 WHITE_LEVEL WHITE_TRIM 100 kΩ
RV6101 WHITE_TRIM WHITE_LEVEL_SUM WHITE_LEVEL_SUM CCW Wiper CW 10 kΩ
R6112 Cbe877b_DRIVE Cbe877b_ISO_MID 499 Ω
R6113 Cbe877b_ISO_MID WHITE_TIP 499 Ω
J6109 AGND Tip Sleeve WHITE_TIP Tip-normal
R6114 PINK_SIGNAL PINK_LEVEL_SUM 100 kΩ
R6115 PINK_LEVEL PINK_TRIM 100 kΩ
RV6102 PINK_TRIM PINK_LEVEL_SUM PINK_LEVEL_SUM CCW Wiper CW 10 kΩ
R6116 Cdbc5aa_DRIVE Cdbc5aa_ISO_MID 499 Ω
R6117 Cdbc5aa_ISO_MID PINK_TIP 499 Ω
J6120 AGND Tip Sleeve PINK_TIP Tip-normal
R6118 BLUE_SHAPE BLUE_LEVEL_SUM 10 kΩ
R6119 BLUE_LEVEL BLUE_TRIM 100 kΩ
RV6103 BLUE_TRIM BLUE_LEVEL_SUM BLUE_LEVEL_SUM CCW Wiper CW 10 kΩ
R6120 Ce44f9a_DRIVE Ce44f9a_ISO_MID 499 Ω
R6121 Ce44f9a_ISO_MID BLUE_TIP 499 Ω
J6110 AGND Tip Sleeve BLUE_TIP Tip-normal
R6122 BROWN_SHAPE BROWN_LEVEL_SUM 20 kΩ
R6123 BROWN_LEVEL BROWN_TRIM 100 kΩ
RV6104 BROWN_TRIM BROWN_LEVEL_SUM BROWN_LEVEL_SUM CCW Wiper CW 10 kΩ
R6124 Cde1183_DRIVE Cde1183_ISO_MID 499 Ω
R6125 Cde1183_ISO_MID BROWN_TIP 499 Ω
J6121 AGND Tip Sleeve BROWN_TIP Tip-normal
U6102A WHITE_AC WHITE_SIGNAL WHITE_SIGNAL OPA4196IDR
U6102B PINK_RAW PINK_SOURCE PINK_SOURCE OPA4196IDR
U6102C PINK_AC PINK_SIGNAL PINK_SIGNAL OPA4196IDR
U6102D AGND BROWN_SUM BROWN_SHAPE OPA4196IDR
U6102E +12V −12V V+ V− OPA4196IDR
U6103A AGND BLUE_SUM BLUE_SHAPE OPA4196IDR
U6103B AGND WHITE_LEVEL_SUM WHITE_LEVEL OPA4196IDR
U6103C AGND PINK_LEVEL_SUM PINK_LEVEL OPA4196IDR
U6103D AGND BLUE_LEVEL_SUM BLUE_LEVEL OPA4196IDR
U6103E +12V −12V V+ V− OPA4196IDR
U6104A AGND BROWN_LEVEL_SUM BROWN_LEVEL OPA4196IDR
U6104B AGND PACK_OPA4196IDR_2_IO_UNUSED2 PACK_OPA4196IDR_2_IO_UNUSED2 OPA4196IDR
U6104C AGND PACK_OPA4196IDR_2_IO_UNUSED3 PACK_OPA4196IDR_2_IO_UNUSED3 OPA4196IDR
U6104D AGND PACK_OPA4196IDR_2_IO_UNUSED4 PACK_OPA4196IDR_2_IO_UNUSED4 OPA4196IDR
U6104E +12V −12V V+ V− OPA4196IDR
U6105A WHITE_LEVEL Cbe877b_DRIVE Cbe877b_DRIVE OPA4196IDR
U6105B PINK_LEVEL Cdbc5aa_DRIVE Cdbc5aa_DRIVE OPA4196IDR
U6105C BLUE_LEVEL Ce44f9a_DRIVE Ce44f9a_DRIVE OPA4196IDR
U6105D BROWN_LEVEL Cde1183_DRIVE Cde1183_DRIVE OPA4196IDR
U6105E +12V −12V V+ V− OPA4196IDR
C6110 NOISE_VDD AGND 100 nF
C6111 +12V AGND 100 nF
C6112 −12V AGND 100 nF
C6113 +12V AGND 100 nF
C6114 −12V AGND 100 nF
C6115 +12V AGND 100 nF
C6116 −12V AGND 100 nF
C6117 +12V AGND 100 nF
C6118 −12V AGND 100 nF

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File: noise.kicad_sch

Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 14/36

Sheet: /E1/
File: envelope.kicad_sch

Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 15/36

Sheet: /E2/
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Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 16/36

Sheet: /E3/
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Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 17/36

Sheet: /E4/
File: envelope_tssop_e4_e5.kicad_sch

Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 18/36

Sheet: /E5/
File: envelope_tssop_e4_e5.kicad_sch

Title:

Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 19/36

Sheet: /E6/
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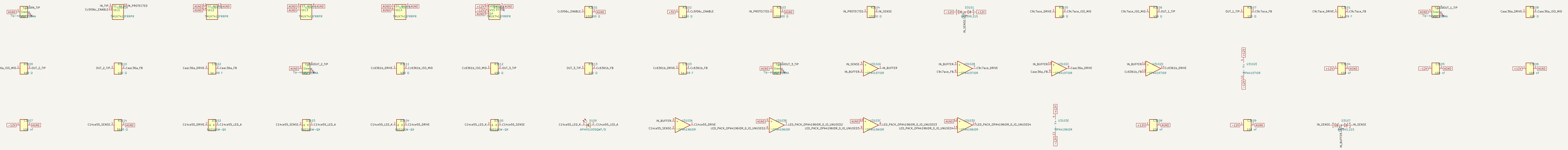
Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 20/36


J109
IN_TIP
AGND
Tip
Sleeve
Tip-normal
U3101A
S1
D1
SEL1
IN_PROTECTED
Cc5f06c_ENABLE
TMUX7412FRRPR
U3101B
S2
D2
SEL2
U3101C
S3
D3
SEL3
U3101D
S4
D4
SEL4
U3101E
VDD
GND
VSS
FF
EP
+12V
-12V
R3101
100000 Ω
R3102
+5V
1000 Ω
R3103
IN_PROTECTED
R3104
10000 Ω
IN_SENSE
D3101
BAV199,215
R3105
C9c7ace_DRIVE
C9c7ace_ISO_MID
499 Ω
R3106
OUT_1_TIP
R3107
C9c7ace_FB
100 Ω
C3101
1e-09 F
J110
R3108
Caac36a_DRIVE
Caac36a_ISO_MID
R3109
OUT_2_TIP
R3110
Caac36a_FB
C3102
J111
R3111
Cc6361b_DRIVE
Cc6361b_ISO_MID
R3112
OUT_3_TIP
R3113
Cc6361b_FB
C3103
J112
U3102A
IN_BUFFER
OPA4197IDR
U3102B
U3102C
U3102D
U3102E
V+
V-
C3104
100 nF
C3105
C3106
C3107
R3114
C14ce55_SENSE
5600 Ω
D3102
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3103
D3104
C14ce55_LED_K
D3105
D109
APHHS1005QWF/D
U3103A
OPA4196IDR
U3103B
LED_PACK_OPA4196IDR_0_IO_UNUSED2
U3103C
LED_PACK_OPA4196IDR_0_IO_UNUSED3
U3103D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3103E
C3108
C3109
D3107
BAT54S,215

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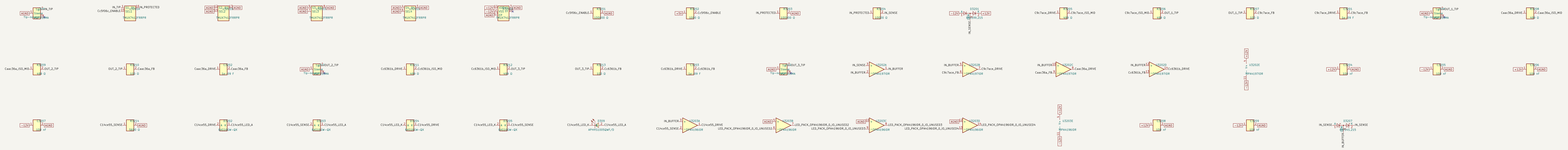
File: mult.kicad_sch

Title:

Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 21/36


J309
AGND
S
Tip
Sleeve
T
IN_TIP
TN
Tip−normal
WQP518MA
U3201A
IN_TIP 1 S1
Cc5f06c_ENABLE 15 SEL1
D1 16 IN_PROTECTED
TMUX7412FRRPR
U3201B
AGND 12 S2
AGND 14 SEL2
D2 13 AGND
TMUX7412FRRPR
U3201C
AGND 9 S3
AGND 7 SEL3
D3 8 AGND
TMUX7412FRRPR
U3201D
AGND 4 S4
AGND 6 SEL4
D4 5 AGND
TMUX7412FRRPR
U3201E
+12V 11 VDD
−12V 2 VSS
AGND 17 EP
GND 3 AGND
FF 10
TMUX7412FRRPR
R3201
Cc5f06c_ENABLE 1
2 AGND
100000 Ω
R3202
+5V 1
2 Cc5f06c_ENABLE
1000 Ω
R3203
IN_PROTECTED 1
2 AGND
100000 Ω
R3204
IN_PROTECTED 1
2 IN_SENSE
10000 Ω
D3201
−12V A 1
K 2 +12V
COM 3
IN_SENSE
BAV199,215
R3205
C9c7ace_DRIVE 1
2 C9c7ace_ISO_MID
499 Ω
R3206
C9c7ace_ISO_MID 1
2 OUT_1_TIP
499 Ω
R3207
OUT_1_TIP 1
2 C9c7ace_FB
100 Ω
C3201
C9c7ace_DRIVE 1
2 C9c7ace_FB
1e−09 F
J409
AGND
S
Tip
Sleeve
T
OUT_1_TIP
TN
Tip−normal
WQP518MA
R3208
Caac36a_DRIVE 1
2 Caac36a_ISO_MID
499 Ω
R3209
Caac36a_ISO_MID 1
2 OUT_2_TIP
499 Ω
R3210
OUT_2_TIP 1
2 Caac36a_FB
100 Ω
C3202
Caac36a_DRIVE 1
2 Caac36a_FB
1e−09 F
J310
AGND
S
Tip
Sleeve
T
OUT_2_TIP
TN
Tip−normal
WQP518MA
R3211
Cc6361b_DRIVE 1
2 Cc6361b_ISO_MID
499 Ω
R3212
Cc6361b_ISO_MID 1
2 OUT_3_TIP
499 Ω
R3213
OUT_3_TIP 1
2 Cc6361b_FB
100 Ω
C3203
Cc6361b_DRIVE 1
2 Cc6361b_FB
1e−09 F
J410
AGND
S
Tip
Sleeve
T
OUT_3_TIP
TN
Tip−normal
WQP518MA
U3202A
IN_SENSE 3 +
IN_BUFFER 2 −
1 IN_BUFFER
OPA4197IDR
U3202B
IN_BUFFER 5 +
C9c7ace_FB 6 −
7 C9c7ace_DRIVE
OPA4197IDR
U3202C
IN_BUFFER 10 +
Caac36a_FB 9 −
8 Caac36a_DRIVE
OPA4197IDR
U3202D
IN_BUFFER 12 +
Cc6361b_FB 13 −
14 Cc6361b_DRIVE
OPA4197IDR
U3202E
+12V 4 V+
−12V 11 V−
OPA4197IDR
C3204
+12V 1
2 AGND
100 nF
C3205
−12V 1
2 AGND
100 nF
C3206
+12V 1
2 AGND
100 nF
C3207
−12V 1
2 AGND
100 nF
R3214
C14ce55_SENSE 1
2 AGND
5600 Ω
D3202
C14ce55_DRIVE 2 A
K 1 C14ce55_LED_A
BAS16GW−QX
D3203
C14ce55_SENSE 2 A
K 1 C14ce55_LED_A
BAS16GW−QX
D3204
C14ce55_LED_K 2 A
K 1 C14ce55_DRIVE
BAS16GW−QX
D3205
C14ce55_LED_K 2 A
K 1 C14ce55_SENSE
BAS16GW−QX
D309
C14ce55_LED_K 1
2 C14ce55_LED_A
APHHS1005QWF/D
U3203A
IN_BUFFER 3 +
C14ce55_SENSE 2 −
1 C14ce55_DRIVE
OPA4196IDR
U3203B
AGND 5 +
LED_PACK_OPA4196IDR_0_IO_UNUSED2 6 −
7 LED_PACK_OPA4196IDR_0_IO_UNUSED2
OPA4196IDR
U3203C
AGND 10 +
LED_PACK_OPA4196IDR_0_IO_UNUSED3 9 −
8 LED_PACK_OPA4196IDR_0_IO_UNUSED3
OPA4196IDR
U3203D
AGND 12 +
LED_PACK_OPA4196IDR_0_IO_UNUSED4 13 −
14 LED_PACK_OPA4196IDR_0_IO_UNUSED4
OPA4196IDR
U3203E
+12V 4 V+
−12V 11 V−
OPA4196IDR
C3208
+12V 1
2 AGND
100 nF
C3209
−12V 1
2 AGND
100 nF
D3207
IN_SENSE A 1
K 2 IN_SENSE
COM
IN_BUFFER
BAT54S,215

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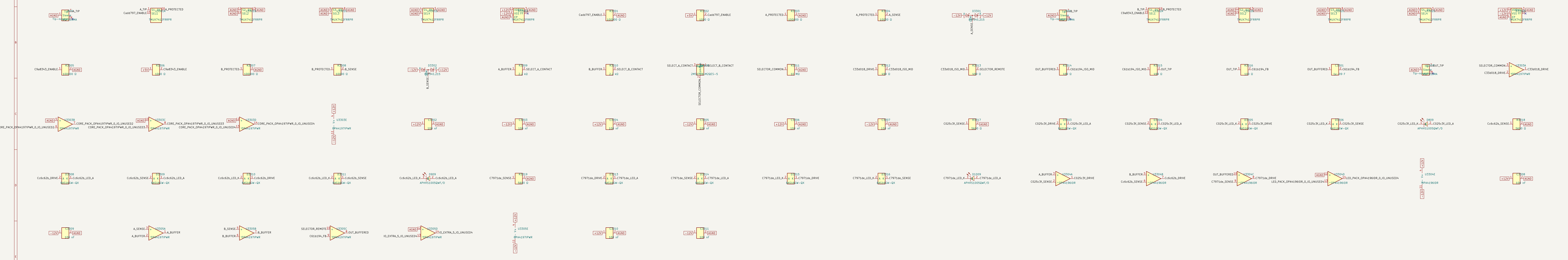
AGND J3309 A_TIP Tip Sleeve Tip-normal TN MJ-3518MA
A_TIP Cadd797_ENABLE S1 D1 SEL1 U3301A A_PROTECTED TMUX7412FRRPR
AGND AGND S2 D2 SEL2 U3301B AGND TMUX7412FRRPR
AGND AGND S3 D3 SEL3 U3301C AGND TMUX7412FRRPR
AGND AGND S4 D4 SEL4 U3301D AGND TMUX7412FRRPR
+12V −12V AGND VDD GND VSS FF EP U3301E AGND TMUX7412FRRPR
Cadd797_ENABLE R3301 AGND 100000 Ω
+5V R3302 Cadd797_ENABLE 1000 Ω
A_PROTECTED R3303 AGND 100000 Ω
A_PROTECTED R3304 A_SENSE 10000 Ω
−12V D3301 +12V A_SENSE BAT54S,215
AGND J3310 B_TIP Tip Sleeve Tip-normal TN MJ-3518MA
B_TIP C9a8343_ENABLE S1 D1 SEL1 U3302A B_PROTECTED TMUX7412FRRPR
AGND AGND S2 D2 SEL2 U3302B AGND TMUX7412FRRPR
AGND AGND S3 D3 SEL3 U3302C AGND TMUX7412FRRPR
AGND AGND S4 D4 SEL4 U3302D AGND TMUX7412FRRPR
+12V −12V AGND VDD GND VSS FF EP U3302E AGND TMUX7412FRRPR
C9a8343_ENABLE R3305 AGND 100000 Ω
+5V R3306 C9a8343_ENABLE 1000 Ω
B_PROTECTED R3307 AGND 100000 Ω
B_PROTECTED R3308 B_SENSE 10000 Ω
−12V D3302 +12V B_SENSE BAT54S,215
A_BUFFER R3309 SELECT_A_CONTACT 2.2 kΩ
B_BUFFER R3310 SELECT_B_CONTACT 2.2 kΩ
SELECT_A_CONTACT SELECT_B_CONTACT SELECTOR_COMMON 2MS1T1B1M2QES-5
SELECTOR_COMMON R3311 AGND 10 MΩ
C33d018_DRIVE R3312 C33d018_ISO_MID 499 Ω
C33d018_ISO_MID R3313 SELECTOR_REMOTE 499 Ω
OUT_BUFFERED R3314 C61b194_ISO_MID 499 Ω
C61b194_ISO_MID R3315 OUT_TIP 499 Ω
OUT_TIP R3316 C61b194_FB 100 Ω
OUT_BUFFERED C3301 C61b194_FB 1e-09 F
AGND J3311 OUT_TIP Tip Sleeve Tip-normal TN MJ-3518MA
SELECTOR_COMMON C33d018_DRIVE U3303A C33d018_DRIVE OPA4197IPWR
AGND CORE_PACK_OPA4197IPWR_0_IO_UNUSED2 U3303B CORE_PACK_OPA4197IPWR_0_IO_UNUSED2 OPA4197IPWR
AGND CORE_PACK_OPA4197IPWR_0_IO_UNUSED3 U3303C CORE_PACK_OPA4197IPWR_0_IO_UNUSED3 OPA4197IPWR
AGND CORE_PACK_OPA4197IPWR_0_IO_UNUSED4 U3303D CORE_PACK_OPA4197IPWR_0_IO_UNUSED4 OPA4197IPWR
+12V −12V V+ V− U3303E OPA4197IPWR
+12V C3302 AGND 100 nF
−12V C3303 AGND 100 nF
+12V C3304 AGND 100 nF
−12V C3305 AGND 100 nF
+12V C3306 AGND 100 nF
−12V C3307 AGND 100 nF
C025c3f_SENSE R3317 AGND 5600 Ω
C025c3f_DRIVE D3303 C025c3f_LED_A BAS16GW-QX
C025c3f_SENSE D3304 C025c3f_LED_A BAS16GW-QX
C025c3f_LED_K D3305 C025c3f_DRIVE BAS16GW-QX
C025c3f_LED_K D3306 C025c3f_SENSE BAS16GW-QX
C025c3f_LED_K D809 C025c3f_LED_A APHHS1005QWF/D
Cc6c62b_SENSE R3318 AGND 5600 Ω
Cc6c62b_DRIVE D3308 Cc6c62b_LED_A BAS16GW-QX
Cc6c62b_SENSE D3309 Cc6c62b_LED_A BAS16GW-QX
Cc6c62b_LED_K D3310 Cc6c62b_DRIVE BAS16GW-QX
Cc6c62b_LED_K D3311 Cc6c62b_SENSE BAS16GW-QX
Cc6c62b_LED_K D909 Cc6c62b_LED_A APHHS1005QWF/D
C7971de_SENSE R3319 AGND 5600 Ω
C7971de_DRIVE D3313 C7971de_LED_A BAS16GW-QX
C7971de_SENSE D3314 C7971de_LED_A BAS16GW-QX
C7971de_LED_K D3315 C7971de_DRIVE BAS16GW-QX
C7971de_LED_K D3316 C7971de_SENSE BAS16GW-QX
C7971de_LED_K D1009 C7971de_LED_A APHHS1005QWF/D
A_BUFFER C025c3f_SENSE U3304A C025c3f_DRIVE OPA4196IDR
B_BUFFER Cc6c62b_SENSE U3304B Cc6c62b_DRIVE OPA4196IDR
OUT_BUFFERED C7971de_SENSE U3304C C7971de_DRIVE OPA4196IDR
AGND LED_PACK_OPA4196IDR_0_IO_UNUSED4 U3304D LED_PACK_OPA4196IDR_0_IO_UNUSED4 OPA4196IDR
+12V −12V V+ V− U3304E OPA4196IDR
+12V C3308 AGND 100 nF
−12V C3309 AGND 100 nF
A_SENSE A_BUFFER U3305A A_BUFFER OPA4197IPWR
B_SENSE B_BUFFER U3305B B_BUFFER OPA4197IPWR
SELECTOR_REMOTE C61b194_FB U3305C OUT_BUFFERED OPA4197IPWR
AGND IO_EXTRA_5_IO_UNUSED4 U3305D IO_EXTRA_5_IO_UNUSED4 OPA4197IPWR
+12V −12V V+ V− U3305E OPA4197IPWR
+12V C3310 AGND 100 nF
−12V C3311 AGND 100 nF

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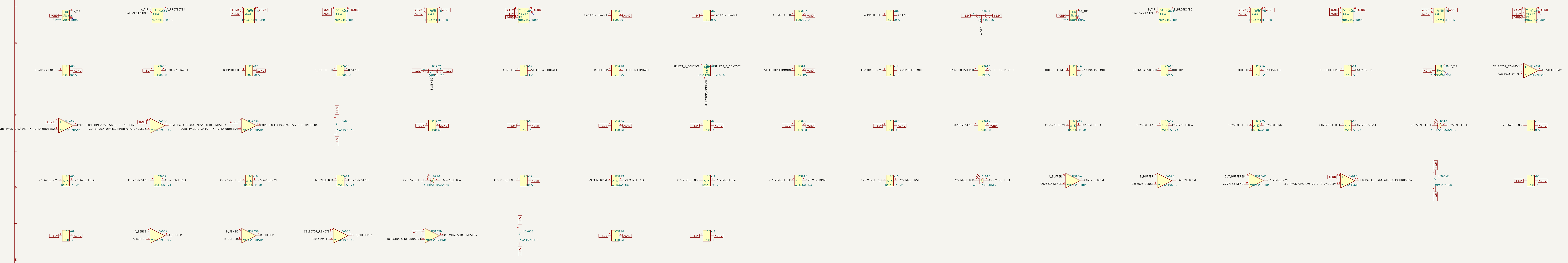
AGND
A_TIP
Tip
Sleeve
Tip-normal
TN
A_TIP
Cadd797_ENABLE
U3401A
SEL1
A_PROTECTED
TMUX7412FRRPR
U3401B
SEL2
U3401C
SEL3
U3401D
SEL4
U3401E
VDD
VSS
EP
GND
+12V
-12V
R3401
Cadd797_ENABLE
100000 Ω
R3402
+5V
1000 Ω
R3403
A_PROTECTED
R3404
A_SENSE
10000 Ω
D3401
BAV94S,215
B_TIP
U3402A
C9a8343_ENABLE
B_PROTECTED
U3402B
U3402C
U3402D
U3402E
R3405
C9a8343_ENABLE
R3406
R3407
B_PROTECTED
R3408
B_SENSE
D3402
R3409
A_BUFFER
SELECT_A_CONTACT
2.2 kΩ
R3410
B_BUFFER
SELECT_B_CONTACT
2MS1T1B1M2QES-5
SELECTOR_COMMON
R3411
10 MΩ
R3412
C33d018_DRIVE
C33d018_ISO_MID
499 Ω
R3413
SELECTOR_REMOTE
R3414
OUT_BUFFERED
C61b194_ISO_MID
R3415
OUT_TIP
R3416
C61b194_FB
100 Ω
C3401
1e-09 F
U3403A
OPA4197IPWR
U3403B
CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
U3403C
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
U3403D
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
U3403E
C3402
100 nF
C3403
C3404
C3405
C3406
C3407
R3417
C025c3f_SENSE
5600 Ω
D3403
C025c3f_DRIVE
C025c3f_LED_A
BAS16GW-QX
D3404
D3405
C025c3f_LED_K
D3406
D810
APHHS1005QWF/D
R3418
Cc6c62b_SENSE
D3408
Cc6c62b_DRIVE
Cc6c62b_LED_A
D3409
D3410
Cc6c62b_LED_K
D3411
D910
R3419
C7971de_SENSE
D3413
C7971de_DRIVE
C7971de_LED_A
D3414
D3415
C7971de_LED_K
D3416
D1010
U3404A
OPA4196IDR
U3404B
U3404C
U3404D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
U3404E
C3408
C3409
U3405A
U3405B
U3405C
U3405D
IO_EXTRA_5_IO_UNUSED4
U3405E
C3410
C3411

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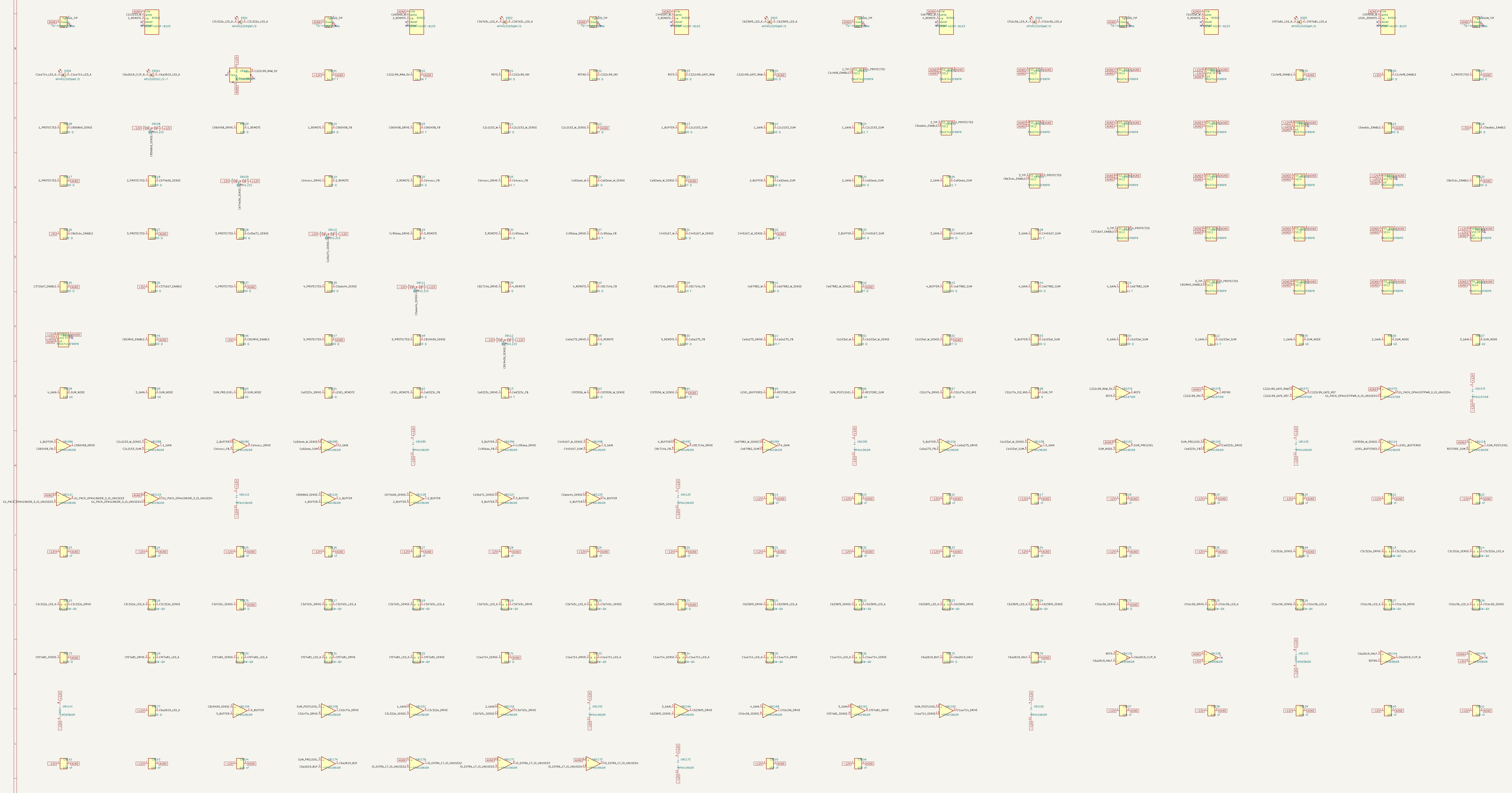

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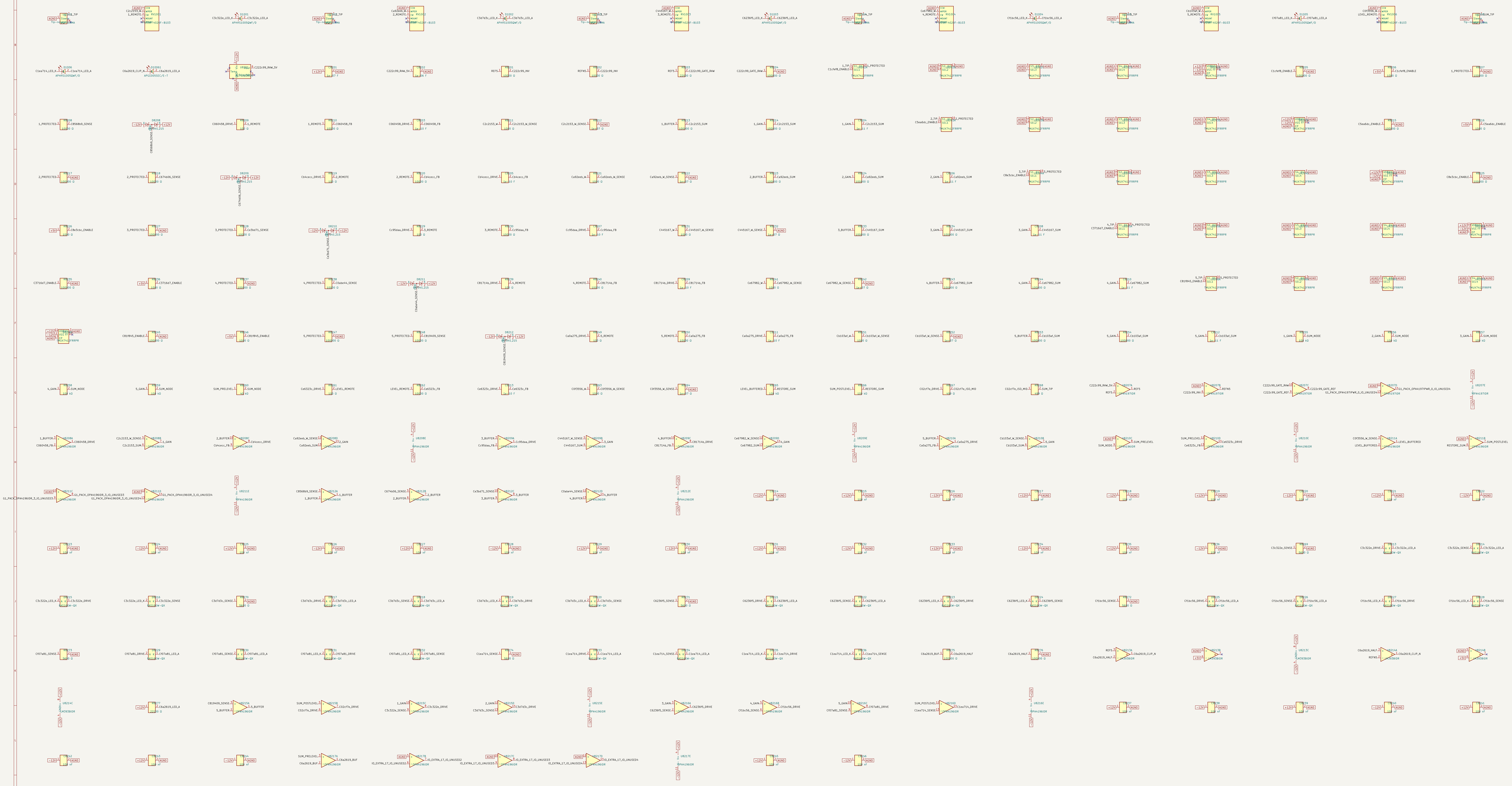

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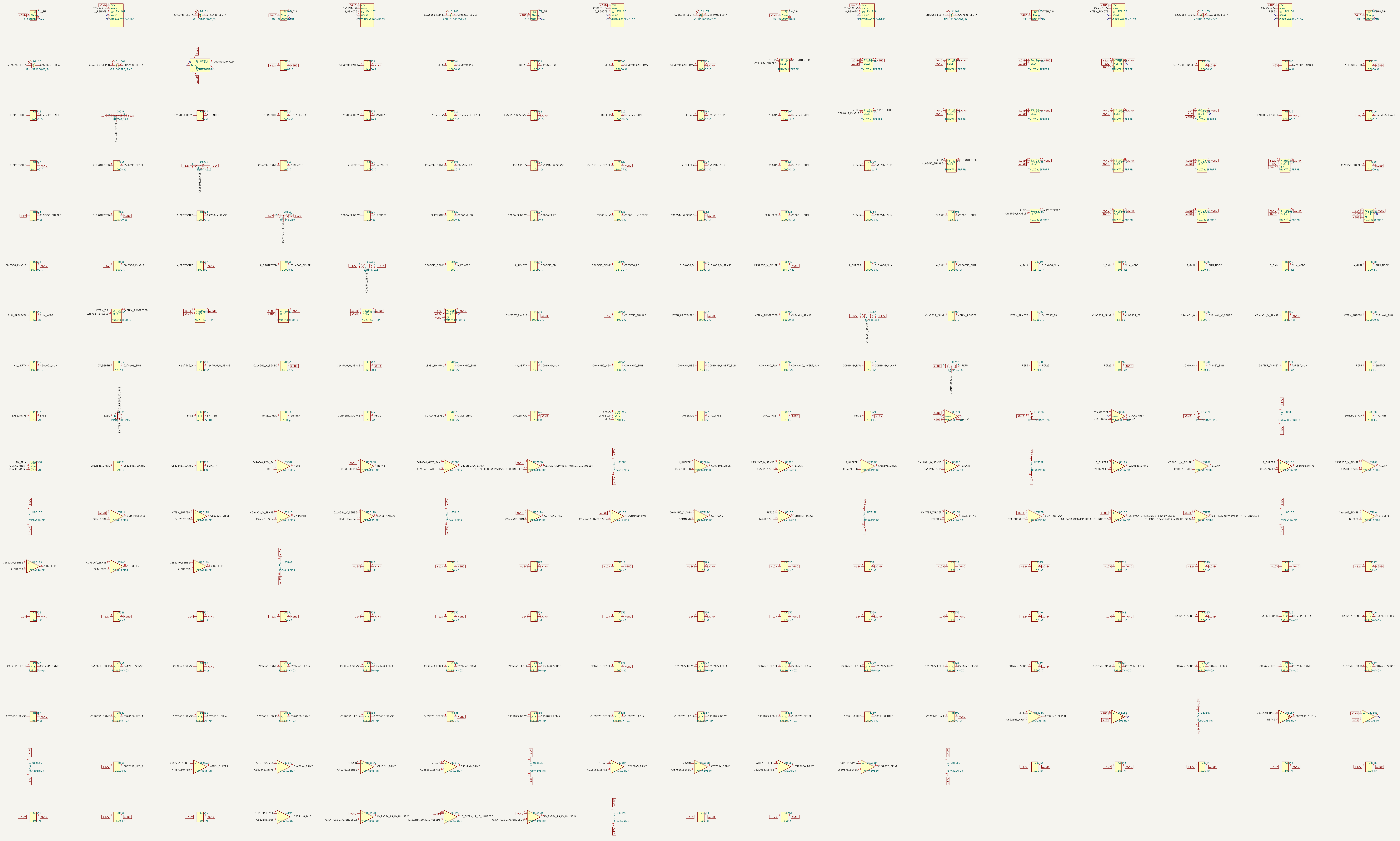

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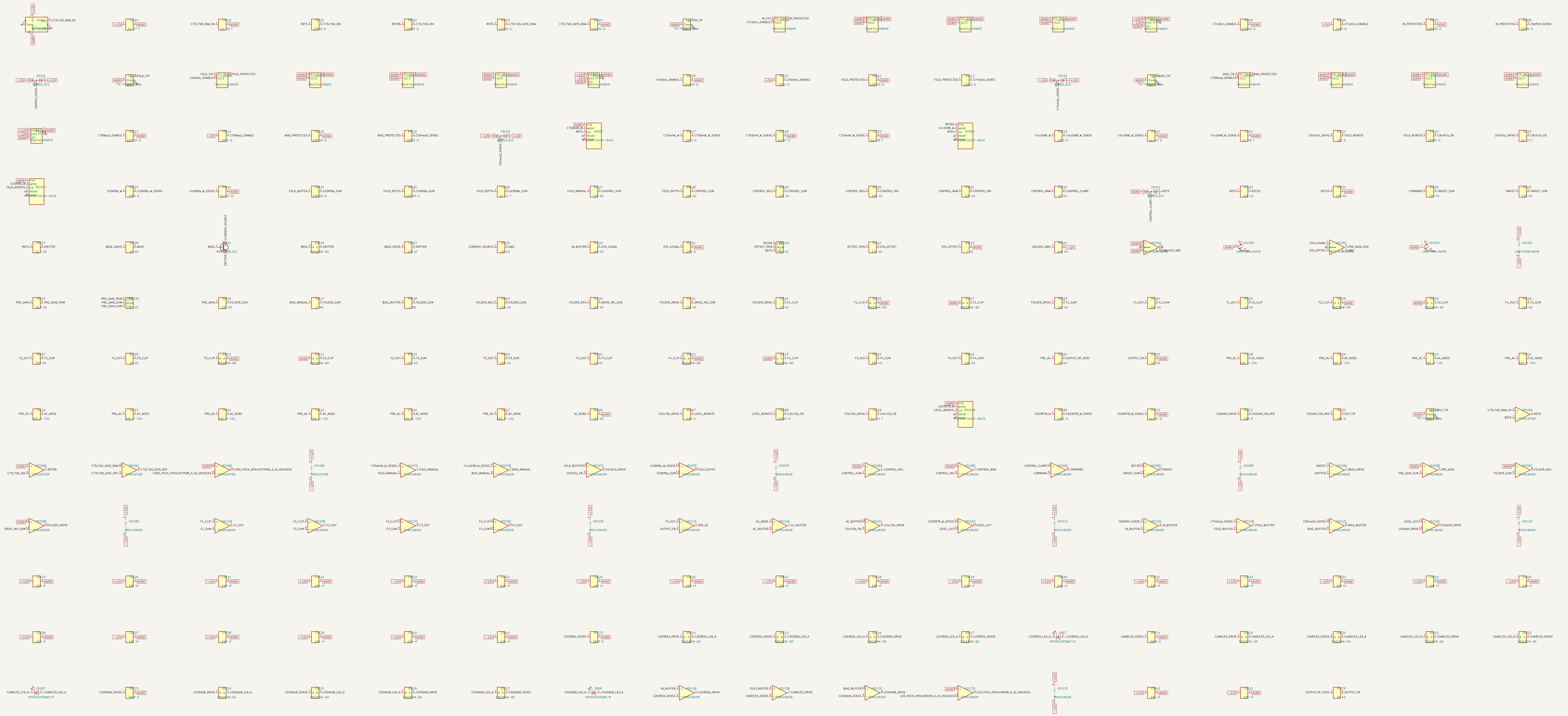

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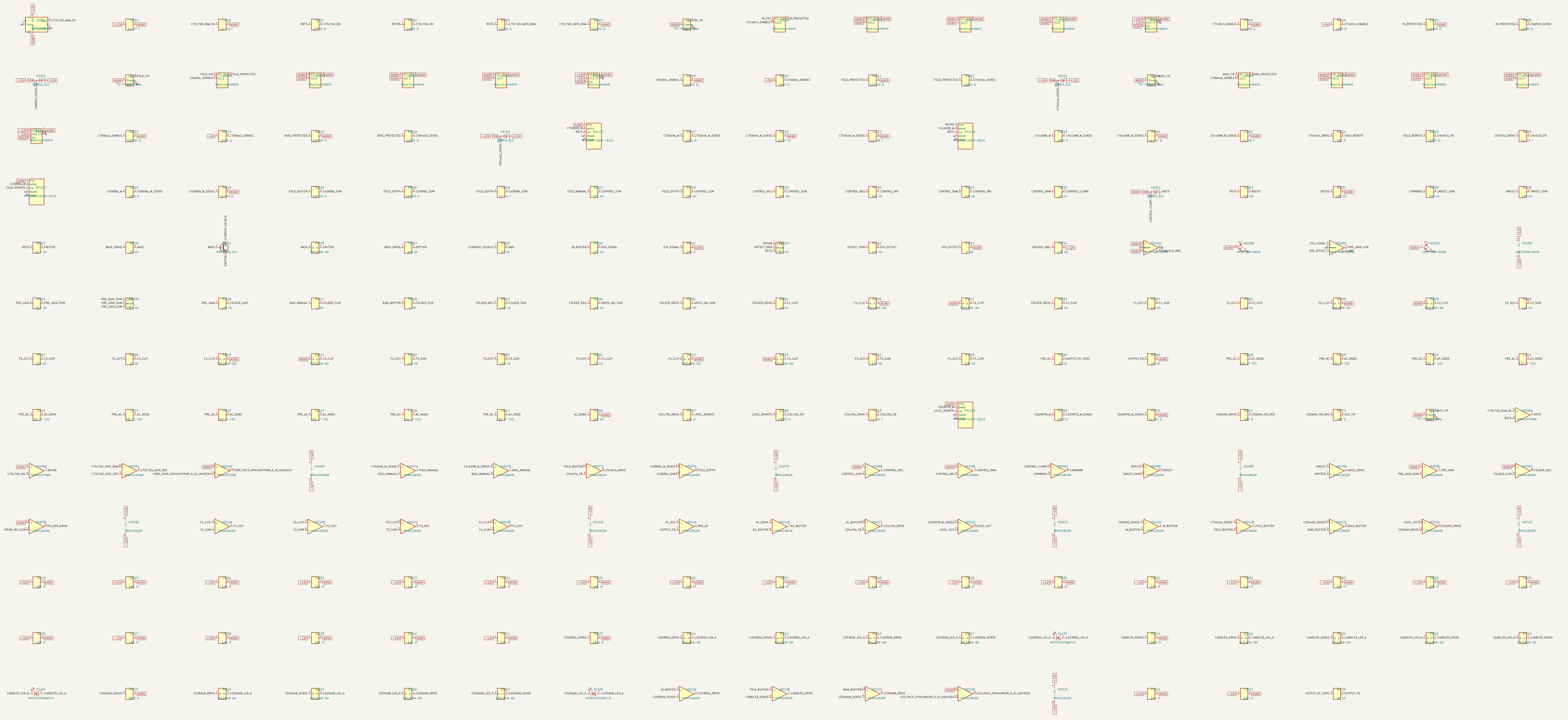

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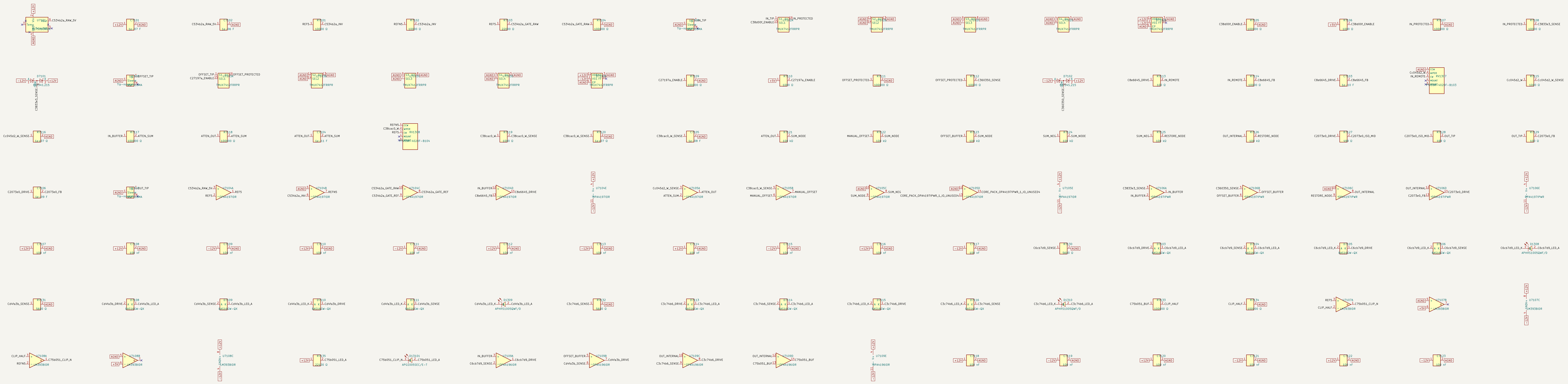

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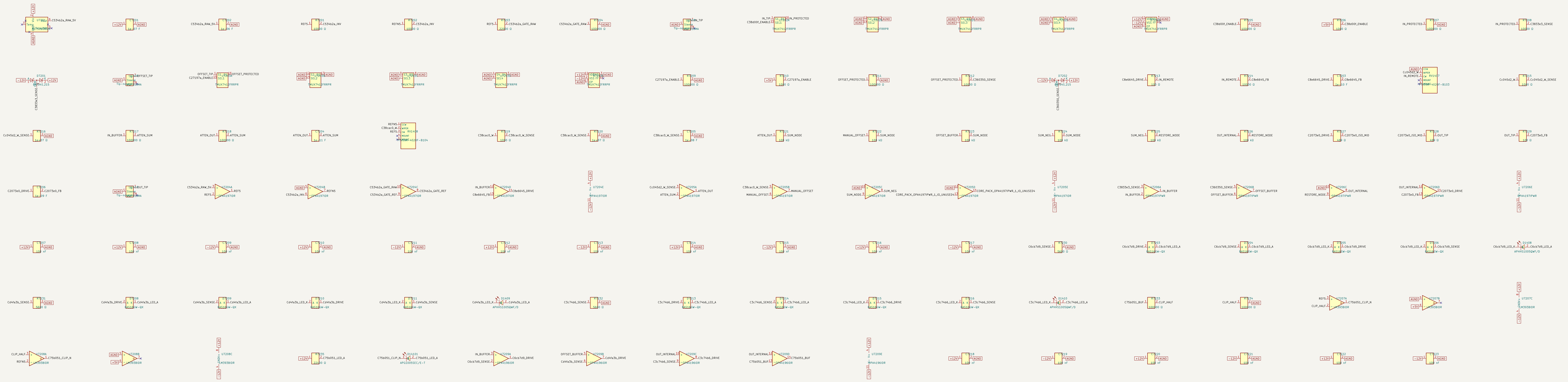
U7201 REF5050AIDR +12V AGND Vin Vout Temp Trim/NR GND C534b2a_RAW_5V
C7201 +12V AGND 1e-07 F
C7202 C534b2a_RAW_5V AGND 1e-06 F
R7201 REF5 C534b2a_INV 10000 Ω
R7202 REFN5 C534b2a_INV 10000 Ω
R7203 REF5 C534b2a_GATE_RAW 22000 Ω
R7204 C534b2a_GATE_RAW AGND 100000 Ω
J1408 AGND IN_TIP Tip Sleeve Tip-normal WQP518MA
U7202A IN_TIP C38d00f_ENABLE IN_PROTECTED S1 D1 SEL1 TMUX7412FRRPR
U7202B AGND S2 D2 SEL2 TMUX7412FRRPR
U7202C AGND S3 D3 SEL3 TMUX7412FRRPR
U7202D AGND S4 D4 SEL4 TMUX7412FRRPR
U7202E +12V −12V AGND VDD GND VSS FF EP TMUX7412FRRPR
R7205 C38d00f_ENABLE AGND 100000 Ω
R7206 +5V C38d00f_ENABLE 1000 Ω
R7207 IN_PROTECTED AGND 100000 Ω
R7208 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7201 −12V +12V BAT54S,215 C3833e3_SENSE
J1409 AGND OFFSET_TIP Tip Sleeve Tip-normal WQP518MA
U7203A OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED SEL1 TMUX7412FRRPR
U7203B AGND SEL2 TMUX7412FRRPR
U7203C AGND SEL3 TMUX7412FRRPR
U7203D AGND SEL4 TMUX7412FRRPR
U7203E +12V −12V AGND VDD GND VSS FF EP TMUX7412FRRPR
R7209 C27197a_ENABLE AGND 100000 Ω
R7210 +5V C27197a_ENABLE 1000 Ω
R7211 OFFSET_PROTECTED AGND 100000 Ω
R7212 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7202 −12V +12V BAT54S,215 C360350_SENSE
R7213 C8e6645_DRIVE IN_REMOTE 100 Ω
R7214 IN_REMOTE C8e6645_FB 10000 Ω
C7203 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1407 AGND Cc045d2_W IN_REMOTE CCW WIPER CW MOUNT PTV09A-4020F-B103
R7215 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7216 Cc045d2_W_SENSE AGND 1e+07 Ω
R7217 IN_BUFFER ATTEN_SUM 100000 Ω
R7218 ATTEN_OUT ATTEN_SUM 100000 Ω
C7204 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1408 REFN5 C38cac0_W REF5 CCW WIPER CW MOUNT PTV09A-4020F-B104
R7219 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7220 C38cac0_W_SENSE AGND 1e+07 Ω
C7205 C38cac0_W_SENSE AGND 1e-08 F
R7221 ATTEN_OUT SUM_NODE 100 kΩ
R7222 MANUAL_OFFSET SUM_NODE 100 kΩ
R7223 OFFSET_BUFFER SUM_NODE 100 kΩ
R7224 SUM_NEG SUM_NODE 100 kΩ
R7225 SUM_NEG RESTORE_NODE 100 kΩ
R7226 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7227 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7228 C2073e0_ISO_MID OUT_TIP 499 Ω
R7229 OUT_TIP C2073e0_FB 100 Ω
C7206 C2073e0_DRIVE C2073e0_FB 1e-09 F
J1410 AGND OUT_TIP Tip Sleeve Tip-normal WQP518MA
U7204A C534b2a_RAW_5V REF5 REF5 OPA4197IDR
U7204B AGND C534b2a_INV REFN5 OPA4197IDR
U7204C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IDR
U7204D IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IDR
U7204E +12V −12V OPA4197IDR
U7205A Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IDR
U7205B C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IDR
U7205C AGND SUM_NODE SUM_NEG OPA4197IDR
U7205D AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 OPA4197IDR
U7205E +12V −12V OPA4197IDR
U7206A C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7206B C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7206C AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7206D OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7206E +12V −12V OPA4197IPWR
C7207 +12V AGND 100 nF
C7208 +12V AGND 100 nF
C7209 −12V AGND 100 nF
C7210 +12V AGND 100 nF
C7211 −12V AGND 100 nF
C7212 +12V AGND 100 nF
C7213 −12V AGND 100 nF
C7214 +12V AGND 100 nF
C7215 −12V AGND 100 nF
C7216 +12V AGND 100 nF
C7217 −12V AGND 100 nF
R7230 C6cb7d9_SENSE AGND 5600 Ω
D7203 C6cb7d9_DRIVE C6cb7d9_LED_A BAS16GW-QX
D7204 C6cb7d9_SENSE C6cb7d9_LED_A BAS16GW-QX
D7205 C6cb7d9_LED_K C6cb7d9_DRIVE BAS16GW-QX
D7206 C6cb7d9_LED_K C6cb7d9_SENSE BAS16GW-QX
D1408 C6cb7d9_LED_K C6cb7d9_LED_A APHHS1005QWF/D
R7231 Cd4fa3b_SENSE AGND 5600 Ω
D7208 Cd4fa3b_DRIVE Cd4fa3b_LED_A BAS16GW-QX
D7209 Cd4fa3b_SENSE Cd4fa3b_LED_A BAS16GW-QX
D7210 Cd4fa3b_LED_K Cd4fa3b_DRIVE BAS16GW-QX
D7211 Cd4fa3b_LED_K Cd4fa3b_SENSE BAS16GW-QX
D1409 Cd4fa3b_LED_K Cd4fa3b_LED_A APHHS1005QWF/D
R7232 C3c74b6_SENSE AGND 5600 Ω
D7213 C3c74b6_DRIVE C3c74b6_LED_A BAS16GW-QX
D7214 C3c74b6_SENSE C3c74b6_LED_A BAS16GW-QX
D7215 C3c74b6_LED_K C3c74b6_DRIVE BAS16GW-QX
D7216 C3c74b6_LED_K C3c74b6_SENSE BAS16GW-QX
D1410 C3c74b6_LED_K C3c74b6_LED_A APHHS1005QWF/D
R7233 C75b051_BUF CLIP_HALF 100000 Ω
R7234 CLIP_HALF AGND 100000 Ω
U7207A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7207B AGND +5V LM393BIDR
U7207C +12V −12V V+ GND LM393BIDR
U7208A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7208B AGND +5V LM393BIDR
U7208C +12V −12V V+ GND LM393BIDR
R7235 +12V C75b051_LED_A 22100 Ω
D14101 C75b051_CLIP_N C75b051_LED_A APG1005SEC/E-T
U7209A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7209B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7209C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7209D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7209E +12V −12V OPA4196IDR
C7218 +12V AGND 100 nF
C7219 −12V AGND 100 nF
C7220 +12V AGND 100 nF
C7221 −12V AGND 100 nF
C7222 +12V AGND 100 nF
C7223 −12V AGND 100 nF

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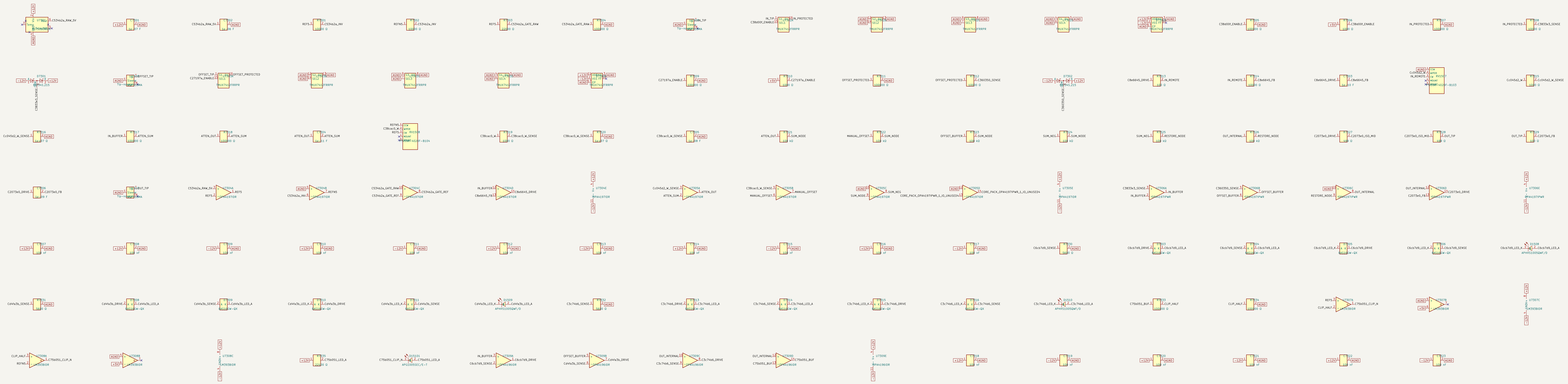

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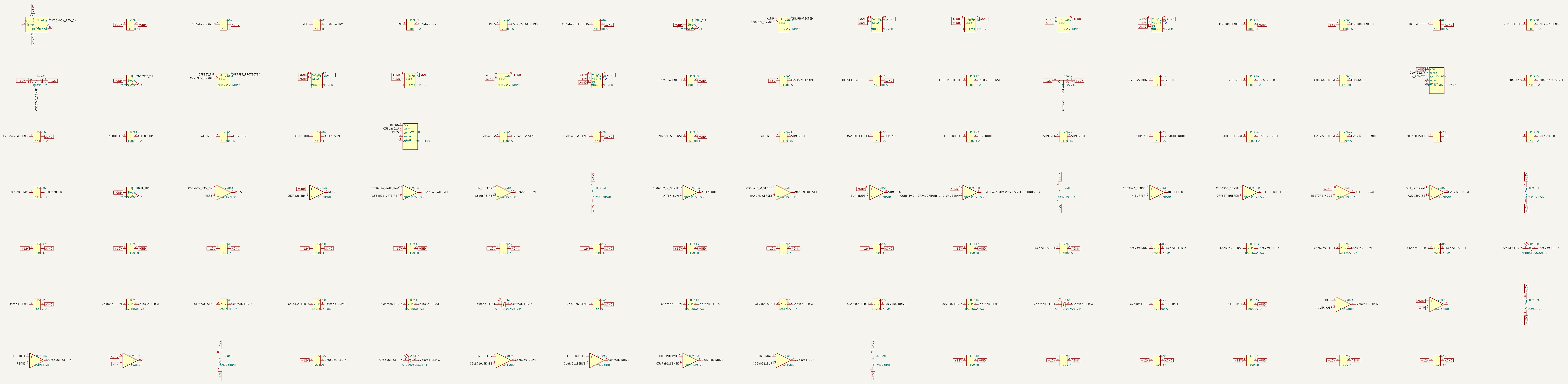

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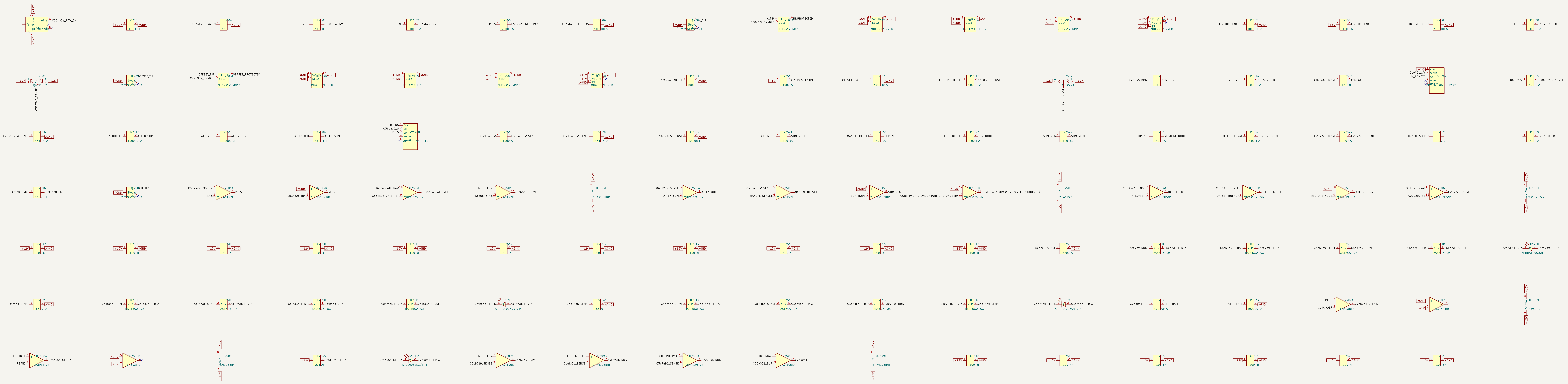
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C7501 +12V AGND 1e-07 F
C7502 C534b2a_RAW_5V AGND 1e-06 F
R7501 REF5 C534b2a_INV 10000 Ω
R7502 REFN5 C534b2a_INV 10000 Ω
R7503 REF5 C534b2a_GATE_RAW 22000 Ω
R7504 C534b2a_GATE_RAW AGND 100000 Ω
J7708 IN_TIP AGND Tip Sleeve Tip-normal
U7502A IN_TIP IN_PROTECTED C38d00f_ENABLE S1 D1 SEL1 TMUX7412FRRPR
U7502B AGND S2 D2 SEL2 TMUX7412FRRPR
U7502C AGND S3 D3 SEL3 TMUX7412FRRPR
U7502D AGND S4 D4 SEL4 TMUX7412FRRPR
U7502E +12V −12V AGND VDD GND VSS FF EP TMUX7412FRRPR
R7505 C38d00f_ENABLE AGND 100000 Ω
R7506 +5V C38d00f_ENABLE 1000 Ω
R7507 IN_PROTECTED AGND 100000 Ω
R7508 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7501 −12V +12V A K BAT54S,215 C3833e3_SENSE
J7709 OFFSET_TIP AGND Tip Sleeve Tip-normal
U7503A OFFSET_TIP OFFSET_PROTECTED C27197a_ENABLE S1 D1 SEL1 TMUX7412FRRPR
U7503B AGND S2 D2 SEL2 TMUX7412FRRPR
U7503C AGND S3 D3 SEL3 TMUX7412FRRPR
U7503D AGND S4 D4 SEL4 TMUX7412FRRPR
U7503E +12V −12V AGND VDD GND VSS FF EP TMUX7412FRRPR
R7509 C27197a_ENABLE AGND 100000 Ω
R7510 +5V C27197a_ENABLE 1000 Ω
R7511 OFFSET_PROTECTED AGND 100000 Ω
R7512 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7502 −12V +12V A K BAT54S,215 C360350_SENSE
R7513 C8e6645_DRIVE IN_REMOTE 100 Ω
R7514 IN_REMOTE C8e6645_FB 10000 Ω
C7503 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1707 AGND Cc045d2_W IN_REMOTE CCW WIPER CW MOUNT PTV09A-4020F-B103
R7515 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7516 Cc045d2_W_SENSE AGND 1e+07 Ω
R7517 IN_BUFFER ATTEN_SUM 100000 Ω
R7518 ATTEN_OUT ATTEN_SUM 100000 Ω
C7504 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1708 REFN5 C38cac0_W REF5 CCW WIPER CW MOUNT PTV09A-4020F-B104
R7519 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7520 C38cac0_W_SENSE AGND 1e+07 Ω
C7505 C38cac0_W_SENSE AGND 1e-08 F
R7521 ATTEN_OUT SUM_NODE 100 kΩ
R7522 MANUAL_OFFSET SUM_NODE 100 kΩ
R7523 OFFSET_BUFFER SUM_NODE 100 kΩ
R7524 SUM_NEG SUM_NODE 100 kΩ
R7525 SUM_NEG RESTORE_NODE 100 kΩ
R7526 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7527 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7528 C2073e0_ISO_MID OUT_TIP 499 Ω
R7529 OUT_TIP C2073e0_FB 100 Ω
C7506 C2073e0_DRIVE C2073e0_FB 1e-09 F
J7710 OUT_TIP AGND Tip Sleeve Tip-normal
U7504A C534b2a_RAW_5V REF5 REF5 OPA4197IDR
U7504B AGND C534b2a_INV REFN5 OPA4197IDR
U7504C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IDR
U7504D IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IDR
U7504E +12V −12V V+ V− OPA4197IDR
U7505A Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IDR
U7505B C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IDR
U7505C AGND SUM_NODE SUM_NEG OPA4197IDR
U7505D AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 OPA4197IDR
U7505E +12V −12V V+ V− OPA4197IDR
U7506A C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7506B C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7506C AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7506D OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7506E +12V −12V V+ V− OPA4197IPWR
C7507 +12V AGND 100 nF
C7508 +12V AGND 100 nF
C7509 −12V AGND 100 nF
C7510 +12V AGND 100 nF
C7511 −12V AGND 100 nF
C7512 +12V AGND 100 nF
C7513 −12V AGND 100 nF
C7514 +12V AGND 100 nF
C7515 −12V AGND 100 nF
C7516 +12V AGND 100 nF
C7517 −12V AGND 100 nF
R7530 C6cb7d9_SENSE AGND 5600 Ω
D7503 C6cb7d9_DRIVE C6cb7d9_LED_A BAS16GW-QX
D7504 C6cb7d9_SENSE C6cb7d9_LED_A BAS16GW-QX
D7505 C6cb7d9_LED_K C6cb7d9_DRIVE BAS16GW-QX
D7506 C6cb7d9_LED_K C6cb7d9_SENSE BAS16GW-QX
D1708 C6cb7d9_LED_K C6cb7d9_LED_A APHHS1005QWF/D
R7531 Cd4fa3b_SENSE AGND 5600 Ω
D7508 Cd4fa3b_DRIVE Cd4fa3b_LED_A BAS16GW-QX
D7509 Cd4fa3b_SENSE Cd4fa3b_LED_A BAS16GW-QX
D7510 Cd4fa3b_LED_K Cd4fa3b_DRIVE BAS16GW-QX
D7511 Cd4fa3b_LED_K Cd4fa3b_SENSE BAS16GW-QX
D1709 Cd4fa3b_LED_K Cd4fa3b_LED_A APHHS1005QWF/D
R7532 C3c74b6_SENSE AGND 5600 Ω
D7513 C3c74b6_DRIVE C3c74b6_LED_A BAS16GW-QX
D7514 C3c74b6_SENSE C3c74b6_LED_A BAS16GW-QX
D7515 C3c74b6_LED_K C3c74b6_DRIVE BAS16GW-QX
D7516 C3c74b6_LED_K C3c74b6_SENSE BAS16GW-QX
D1710 C3c74b6_LED_K C3c74b6_LED_A APHHS1005QWF/D
R7533 C75b051_BUF CLIP_HALF 100000 Ω
R7534 CLIP_HALF AGND 100000 Ω
U7507A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7507B AGND +5V LM393BIDR
U7507C +12V −12V V+ GND LM393BIDR
U7508A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7508B AGND +5V LM393BIDR
U7508C +12V −12V V+ GND LM393BIDR
R7535 +12V C75b051_LED_A 22100 Ω
D17101 C75b051_CLIP_N C75b051_LED_A APG1005SEC/E-T
U7509A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7509B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7509C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7509D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7509E +12V −12V V+ V− OPA4196IDR
C7518 +12V AGND 100 nF
C7519 −12V AGND 100 nF
C7520 +12V AGND 100 nF
C7521 −12V AGND 100 nF
C7522 +12V AGND 100 nF
C7523 −12V AGND 100 nF

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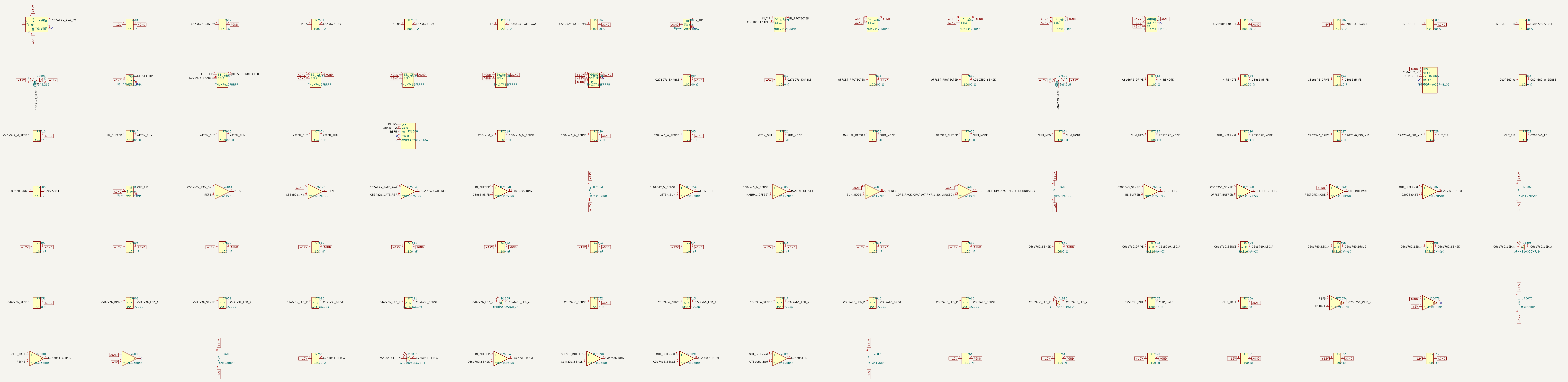
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C7602 1e-06 F C534b2a_RAW_5V AGND
R7601 10000 Ω REF5 C534b2a_INV
R7602 10000 Ω REFN5 C534b2a_INV
R7603 22000 Ω REF5 C534b2a_GATE_RAW
R7604 100000 Ω C534b2a_GATE_RAW AGND
J1808 Tip Sleeve Tip-normal IN_TIP AGND
U7602A TMUX7412FRRPR S1 D1 SEL1 IN_TIP C38d00f_ENABLE IN_PROTECTED
U7602B TMUX7412FRRPR S2 D2 SEL2 AGND
U7602C TMUX7412FRRPR S3 D3 SEL3 AGND
U7602D TMUX7412FRRPR S4 D4 SEL4 AGND
U7602E TMUX7412FRRPR VDD GND VSS FF EP +12V −12V AGND
R7605 100000 Ω C38d00f_ENABLE AGND
R7606 1000 Ω +5V C38d00f_ENABLE
R7607 100000 Ω IN_PROTECTED AGND
R7608 10000 Ω IN_PROTECTED C3833e3_SENSE
D7601 BAT54S,215 A K COM −12V +12V C3833e3_SENSE
J1809 Tip Sleeve Tip-normal OFFSET_TIP AGND
U7603A TMUX7412FRRPR S1 D1 SEL1 OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED
U7603B TMUX7412FRRPR S2 D2 SEL2 AGND
U7603C TMUX7412FRRPR S3 D3 SEL3 AGND
U7603D TMUX7412FRRPR S4 D4 SEL4 AGND
U7603E TMUX7412FRRPR VDD GND VSS FF EP +12V −12V AGND
R7609 100000 Ω C27197a_ENABLE AGND
R7610 1000 Ω +5V C27197a_ENABLE
R7611 100000 Ω OFFSET_PROTECTED AGND
R7612 10000 Ω OFFSET_PROTECTED C360350_SENSE
D7602 BAT54S,215 A K COM −12V +12V C360350_SENSE
R7613 100 Ω C8e6645_DRIVE IN_REMOTE
R7614 10000 Ω IN_REMOTE C8e6645_FB
C7603 1e-10 F C8e6645_DRIVE C8e6645_FB
RV1807 PTV09A-4020F-B103 CCW WIPER CW MOUNT MOUNT AGND Cc045d2_W IN_REMOTE
R7615 1000 Ω Cc045d2_W Cc045d2_W_SENSE
R7616 1e+07 Ω Cc045d2_W_SENSE AGND
R7617 100000 Ω IN_BUFFER ATTEN_SUM
R7618 100000 Ω ATTEN_OUT ATTEN_SUM
C7604 1e-11 F ATTEN_OUT ATTEN_SUM
RV1808 PTV09A-4020F-B104 CCW WIPER CW MOUNT MOUNT REFN5 C38cac0_W REF5
R7619 1000 Ω C38cac0_W C38cac0_W_SENSE
R7620 1e+07 Ω C38cac0_W_SENSE AGND
C7605 1e-08 F C38cac0_W_SENSE AGND
R7621 100 kΩ ATTEN_OUT SUM_NODE
R7622 100 kΩ MANUAL_OFFSET SUM_NODE
R7623 100 kΩ OFFSET_BUFFER SUM_NODE
R7624 100 kΩ SUM_NEG SUM_NODE
R7625 100 kΩ SUM_NEG RESTORE_NODE
R7626 100 kΩ OUT_INTERNAL RESTORE_NODE
R7627 499 Ω C2073e0_DRIVE C2073e0_ISO_MID
R7628 499 Ω C2073e0_ISO_MID OUT_TIP
R7629 100 Ω OUT_TIP C2073e0_FB
C7606 1e-09 F C2073e0_DRIVE C2073e0_FB
J1810 Tip Sleeve Tip-normal OUT_TIP AGND
U7604A OPA4197IDR C534b2a_RAW_5V REF5 REF5
U7604B OPA4197IDR AGND C534b2a_INV REFN5
U7604C OPA4197IDR C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF
U7604D OPA4197IDR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7604E OPA4197IDR V+ V− +12V −12V
U7605A OPA4197IDR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7605B OPA4197IDR C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET
U7605C OPA4197IDR AGND SUM_NODE SUM_NEG
U7605D OPA4197IDR AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4
U7605E OPA4197IDR V+ V− +12V −12V
U7606A OPA4197IPWR C3833e3_SENSE IN_BUFFER IN_BUFFER
U7606B OPA4197IPWR C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER
U7606C OPA4197IPWR AGND RESTORE_NODE OUT_INTERNAL
U7606D OPA4197IPWR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7606E OPA4197IPWR V+ V− +12V −12V
C7607 100 nF +12V AGND
C7608 100 nF +12V AGND
C7609 100 nF −12V AGND
C7610 100 nF +12V AGND
C7611 100 nF −12V AGND
C7612 100 nF +12V AGND
C7613 100 nF −12V AGND
C7614 100 nF +12V AGND
C7615 100 nF −12V AGND
C7616 100 nF +12V AGND
C7617 100 nF −12V AGND
R7630 5600 Ω C6cb7d9_SENSE AGND
D7603 BAS16GW-QX A K C6cb7d9_DRIVE C6cb7d9_LED_A
D7604 BAS16GW-QX A K C6cb7d9_SENSE C6cb7d9_LED_A
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D7606 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_SENSE
D1808 APHHS1005QWF/D C6cb7d9_LED_K C6cb7d9_LED_A
R7631 5600 Ω Cd4fa3b_SENSE AGND
D7608 BAS16GW-QX A K Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7609 BAS16GW-QX A K Cd4fa3b_SENSE Cd4fa3b_LED_A
D7610 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7611 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_SENSE
D1809 APHHS1005QWF/D Cd4fa3b_LED_K Cd4fa3b_LED_A
R7632 5600 Ω C3c74b6_SENSE AGND
D7613 BAS16GW-QX A K C3c74b6_DRIVE C3c74b6_LED_A
D7614 BAS16GW-QX A K C3c74b6_SENSE C3c74b6_LED_A
D7615 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_DRIVE
D7616 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_SENSE
D1810 APHHS1005QWF/D C3c74b6_LED_K C3c74b6_LED_A
R7633 100000 Ω C75b051_BUF CLIP_HALF
R7634 100000 Ω CLIP_HALF AGND
U7607A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7607B LM393BIDR AGND +5V
U7607C LM393BIDR GND V+ +12V −12V
U7608A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7608B LM393BIDR AGND +5V
U7608C LM393BIDR GND V+ +12V −12V
R7635 22100 Ω +12V C75b051_LED_A
D18101 APG1005SEC/E-T C75b051_CLIP_N C75b051_LED_A
U7609A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7609B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7609C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7609D OPA4196IDR OUT_INTERNAL C75b051_BUF C75b051_BUF
U7609E OPA4196IDR V+ V− +12V −12V
C7618 100 nF +12V AGND
C7619 100 nF −12V AGND
C7620 100 nF +12V AGND
C7621 100 nF −12V AGND
C7622 100 nF +12V AGND
C7623 100 nF −12V AGND

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